

ReproBot: A Multi-Agent System for Automated Scientific Paper Replication

Third Progress Report

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Abstract

ReproBot is a multi-agent system that reads a machine learning paper, extracts its claims and experimental setup, generates an implementation, executes it in a sandbox, and repairs execution failures through a bounded retry loop. This report covers the phase in which it produced its first replication measurements. Full training on the project’s CIFAR-10 benchmark would take weeks per paper on the available CPU, so we evaluated on papers whose original experiments run in minutes, and generalized code generation beyond image classification: the Coder now writes a plain PyTorch training loop for neural models and a scikit-learn estimator for classical ones, under a task-agnostic metrics contract. Five papers — a network with a margin-based loss, a tabular regression network, a kernel support vector machine, a random forest and a soft decision tree — were carried through the pipeline and compared against their published claims by hand. Three reproductions land close to their claims: an MNIST network at 0.82% test error against 0.87%, after one unstated hyperparameter was corrected by hand; a support vector machine at 96.63% against 96.9%, whose implementation reproduces the paper’s exact numbers when given the paper’s settings; and a random forest at 87.73% against 87.3%. A housing-price model misses its claim, with a test RMSE of 4.48 against 3.02, by a margin a single data split cannot explain, and a soft decision tree whose paper states little beyond its depth did not learn at all. The retry loop repaired three naturally occurring defects, and six prompt failures observed in real runs became explicit rules. The central finding is that the pipeline reported *success* on two runs that had failed entirely: nothing yet judges whether a reproduced number is plausible, which makes an automated Critic the most important missing component. We also describe the team’s first front end, a dashboard over the pipeline’s outputs.

1 Introduction

Reproducing a machine learning paper’s reported results is mechanical in principle and laborious in practice, and most published work ships no usable implementation. ReproBot automates the process as a pipeline of specialized stages. A vision-language model converts the paper’s PDF into Markdown, reading figures as well as text. A Reader extracts five structured fields — a method summary, architecture notes, the paper’s claimed results, its hyperparameters and its data pipeline — and cross-checks them with a validation loop. A Coder turns that extraction into a self-contained training script. A Runner executes the script inside a Docker container through four escalating stages: **probe**, a few optimizer steps; **smoke**, one epoch on a small slice; **capped**, five epochs on a small slice; and **full**, the paper’s own setup. When a stage fails, a cheap model classifies the failure, and an Orchestrator feeds that diagnosis back to the Coder and retries, within a fixed budget. The planned final stage, a Critic that compares each reproduced number with the paper’s claim under an explicit tolerance, does not yet exist.

At the start of this phase the pipeline could execute generated code end to end, but it had produced no replication result. Every run on its CIFAR-10 target papers had been a short execution check, because a full training run on the available CPU was measured at roughly three weeks per paper. This phase changed that in three ways. Section 2 describes a change of direction: evaluating on papers small enough to train for real on a CPU, and a team decision that ReproBot must handle several kinds of paper and several model families. Section 3 describes the resulting changes to the Coder and Runner, and Section 4 the prompt failures that real runs exposed and the rules that now prevent them. Section 5 reports the measurements, including a training collapse, the ablation that explained it, and five comparisons against published claims. Section 6 describes the team’s pipeline viewer, and Sec-

tions 7–12 collect the engineering findings, methodology, progress against plan, limitations and next steps.

Table 1 summarizes the open problems at the start of the phase and where each now stands.

2 A Change of Direction

2.1 The compute wall

The CIFAR-10 papers chosen as ReproBot’s first targets are the natural benchmark for the proposal’s image-classification scope, and they are unreachable on the hardware available to the project. At the measured throughput of 5.3 training samples per second, one *Wide Residual Networks* run at the paper’s own setup takes approximately 22 days. Every number previously produced for these papers therefore came from short execution checks that train for one epoch on a few hundred images, and could say nothing about fidelity.

Securing GPU compute remains necessary for that benchmark. It is not, however, necessary to test the claim ReproBot actually makes — that a paper can be read, implemented, executed and checked automatically — which requires only a paper whose original experiment is small. We therefore looked for papers whose headline result trains in minutes on a CPU, while leaving the curated CIFAR-10 set untouched.

2.2 More than one kind of paper

The search surfaced a structural problem. The pipeline was, in practice, hardwired to image classification: the Coder’s prompt assumed `torchvision` datasets and the HuggingFace `Trainer` image-model interface, the metrics contract was keyed on accuracy, and the Runner’s image carried no tabular libraries. An in-scope MNIST paper would have tested the pipeline as it stood. A housing-price regression paper could not have run at all.

The team’s decision was that this was a defect rather than a scope boundary: a system that replicates only one task type is not a general replication system, and ReproBot should handle several. CIFAR-10 remains the main evaluation set, but no stage may now assume images. Both candidate papers were retained, precisely because they differ:

- **Tang (2013)**, *Deep Learning using Linear Support Vector Machines* [1]. A classification paper whose contribution is a loss: the softmax output layer is replaced by a linear L2-SVM trained with a squared hinge loss. The targeted claim is 0.87% MNIST test error. The setup — PCA to 70 dimensions, two 512-unit hidden layers, 400 epochs — is stated in unusual detail, which makes the two values it omits all the more consequential (Section 5.3).

- **Wijaya (2023)**, *Multi Level Dense Layer Neural Network Model for Housing Price Prediction* [2]. A tabular regression paper in Keras, on the Boston Housing data [3]: 506 rows, 13 features, a stated 405/101 train/test split, 1000 epochs. The targeted claim is a test RMSE of 3.02. It tests three things CIFAR-10 cannot: a non-image data source, a regression metric where lower is better, and a paper written for a different framework.

A replication system should also handle the models much of applied machine learning still relies on: kernel methods, decision trees and ensembles. Three further papers were therefore chosen to cover model families rather than tasks:

- **Hsu, Chang and Lin (2003)**, *A Practical Guide to Support Vector Classification* [4]. A widely used tutorial whose worked examples report exact accuracies from the LIBSVM tools [5]. The target is its astroparticle-physics example, `svmguide1` (3,089 training and 4,000 test instances, four features): 96.9% test accuracy after scaling each feature to $[-1, 1]$ and choosing the RBF kernel’s C and γ by a cross-validated grid search. Its appendix gives the unrounded value, 96.875%, at $C = \gamma = 2$.
- **Xiao, Rasul and Vollgraf (2017)**, *Fashion-MNIST* [6]. A dataset paper whose benchmark table evaluates about 130 configurations of 13 scikit-learn classifiers [7]. The target is its random forest (100 trees, entropy criterion, maximum depth 100): 0.873 test accuracy, the mean of five runs with shuffled training data.
- **Frosst and Hinton (2017)**, *Distilling a Neural Network Into a Soft Decision Tree* [8]. A binary tree whose inner nodes are learned linear filters, trained by gradient descent. The target is 94.45% MNIST test accuracy for a depth-8 tree trained on true labels. The paper states the depth and little else — no learning rate, batch size, epoch count, penalty strength or temperature — so this target deliberately tests how the pipeline behaves when an experiment is under-specified.

Table 2 summarizes the five targets. All were kept in a separate directory, so the curated CIFAR-10 set remains exactly as its curator left it.

3 Generalizing the Coder and Runner

3.1 Dropping the HuggingFace Trainer

The Coder previously emitted scripts built on HuggingFace’s `Trainer`. Its model contract — image

Table 1: Open problems at the start of this phase, and their status now.

At the start of the phase	Now
No reproduced number had ever been compared with a claim	Five claims measured at full fidelity and compared by hand (Section 5.4); the Critic still does not exist
Full-fidelity runs infeasible (~ 22 days per CIFAR-10 paper)	Unchanged for CIFAR-10; sidestepped with CPU-sized papers whose full runs take between one minute and one hour
Only CIFAR-10 image classifiers, all built with HuggingFace Trainer	Seven papers carried end to end, spanning CNNs, MLPs, a regression network, a kernel SVM, a random forest and a soft decision tree
The retry loop had repaired only a deliberately injected fault	Three naturally occurring defects repaired (Section 5.2)
Four of seven Orchestrator verdicts never observed in a real run	environment_error occurred in a real run — wrongly assigned, and the triage prompt corrected (Section 4); three remain unobserved
Retries regenerate whole scripts rather than patching them	Unchanged, and now measured: one-line fixes produced scripts only 22–27% similar to their predecessors
Reader validation never converged; two CIFAR-10 papers cannot be extracted	Two papers now converge with no flags left; the rest unchanged

Table 2: The five CPU-sized replication targets of this phase.

Paper	Model family	Data	Targeted claim
Tang (2013)	MLP with an L2-SVM output layer	MNIST	0.87% test error
Wijaya (2023)	Multi-branch dense network (Keras)	Boston Housing, tabular	Test RMSE 3.02
Hsu, Chang and Lin (2003)	RBF support vector machine	svmguide1, tabular	96.9% test accuracy
Xiao et al. (2017)	Random forest (scikit-learn)	Fashion-MNIST	0.873 test accuracy, 5 runs
Frosst and Hinton (2017)	Soft decision tree	MNIST	94.45% test accuracy

tensors in, logits out — fits a custom margin loss or a regression head poorly, and that alone argued for replacing it. The deciding reason, however, was that **Trainer** applies defaults that no paper states, and this was demonstrated rather than assumed.

When the generalized Coder regenerated its *Network In Network* script as a regression test, the new plain-PyTorch script kept the paper’s architecture — a genuine `mlpconv` cascade with no fully-connected layer — and passed its execution probe. But its loss reached approximately 947, where the previous **Trainer**-based script had reported 2.30. The cause was a default: the old script never set `max_grad_norm`, and therefore inherited **Trainer**’s gradient clipping at a norm of 1.0, which we confirmed inside the container. The paper uses no clipping. Faithful to the paper and without its unstated initialization, the same learning rate diverges.

This changes how earlier results should be read. The *Network In Network* runs made before this change executed under a stabilizing mechanism the paper does not describe. Their numbers had only ever been treated as execution checks, not replication results, but the scripts that produced them were less faithful to the paper than they appeared. The Coder is now forbidden from using **Trainer**, **accelerate**, **Lightning** or **Keras**, and writes an explicit training loop in which every mechanism is visible.

3.2 Always PyTorch, whatever the paper used

Wijaya’s model was built in Keras. Rather than generate code for a second framework, the Coder always writes PyTorch, and the prompt treats the source framework’s defaults as part of the method: where a paper is silent, it inherited its framework’s behaviour, so a faithful translation must reproduce that behaviour. The prompt names the Keras defaults that differ — Adam’s $\epsilon = 10^{-7}$, Glorot initialization, batch normalization with momentum 0.99 and $\epsilon = 10^{-3}$, and a `fit()` batch size of 32 — and requires every remaining difference to be disclosed. Section 7 reports that this rule was followed only in part.

3.3 A task-agnostic metrics contract

Every generated script writes a fixed-shape metrics file that the Runner parses and a future Critic will read. The original contract was designed for image classification and assumed accuracy. Its generalization is summarized in Table 3; the most consequential addition is **higher_is_better**, since a bare value of 4.0 is meaningless without knowing whether it is an error or an accuracy.

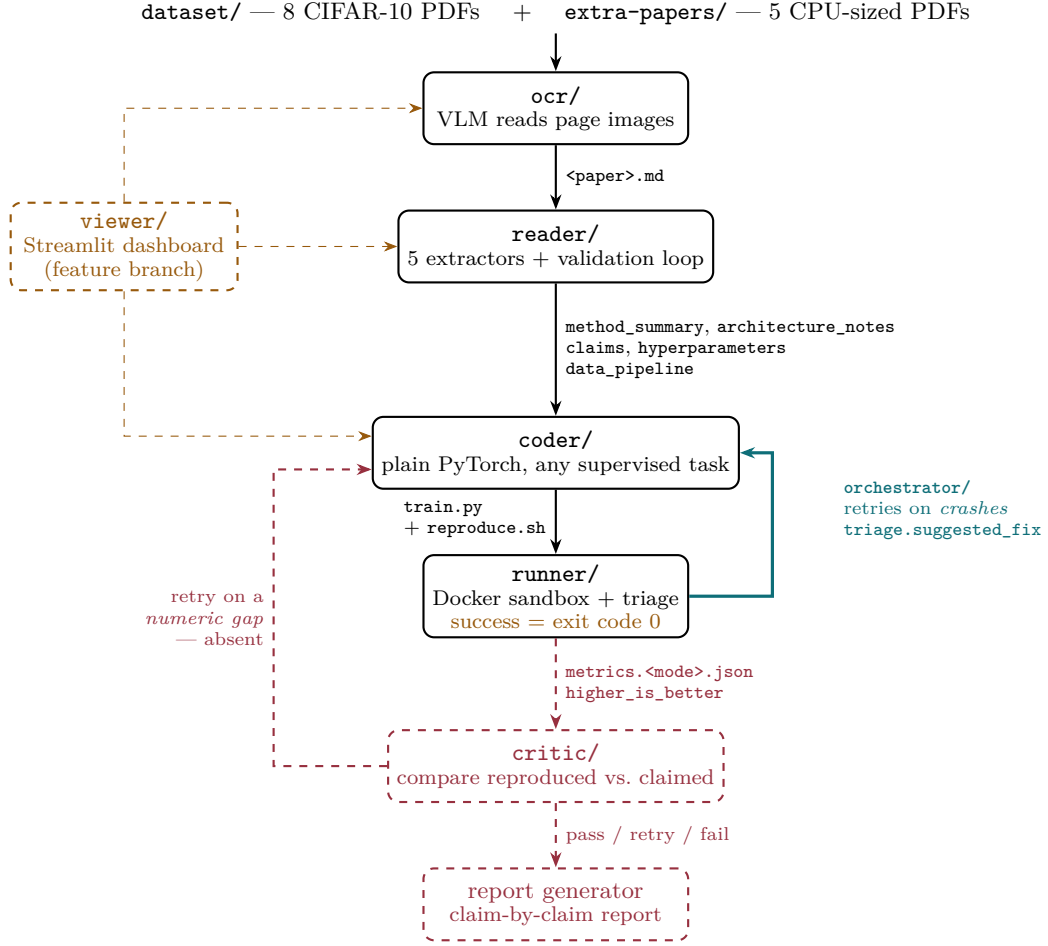


Figure 1: ReproBot on 13.09.2026. Solid components are built and verified by execution; dashed red components do not exist; the dashed amber component is on a feature branch. The pipeline’s shape predates this phase, but its inputs now include CPU-sized papers beyond image classification, and the Coder writes a plain PyTorch loop or a scikit-learn estimator depending on the paper’s model family. The label inside the Runner marks this phase’s central finding: a run is reported as a success when its script exits normally, whatever number it produced.

Table 3: Changes to the metrics contract. **metric** and **unit** are copied verbatim from the targeted claim, so a comparison against the claim needs no conversion. Files in the old shape still parse.

Field	Change
train_metric, eval_metric	Replace train_accuracy and eval_accuracy : the claim’s own metric on each split
higher_is_better	New; the direction of the metric
task_type	New; inferred by the Coder from the claim, e.g. classification or regression
value, metric, unit	Unchanged; the headline number and its claim-verbatim label

3.4 Data sourcing, and verifying what was fetched

Image datasets still come from **torchvision**. Tabular data is fetched from OpenML, pinned by dataset identifier, before any direct URL is considered, and every download lands in the shared dataset cache the Runner mounts. The preference order was not a design choice made in advance; Section 4 describes the failure that produced it.

The generated script must also check, in code and before training, that the dataset it fetched is the one it intended — its name, row count and column count — and raise otherwise. It downloads a different dataset that happens to carry that number, trains on it, and reports a plausible metric for the wrong problem. Where a paper gives split sizes but not how the split was drawn, the script uses a seeded random split and records the assumption.

3.5 Classical estimators in scikit-learn

Writing everything in PyTorch is right for neural networks and wrong for everything else. Reimplementing a support vector machine as a PyTorch model trained by stochastic gradient descent on a hinge loss changes both the optimizer and the solution it reaches; a paper that ran LIBSVM ran a specific quadratic-programming solver. The Coder therefore now chooses its library by model family. Classical estimators — support vector machines, decision trees and forests, k -nearest neighbours, logistic regression, naive Bayes and similar — are built with scikit-learn, whose SVC wraps LIBSVM itself, with every hyperparameter passed explicitly by name. Neural models stay in PyTorch, including tree-shaped models trained by gradient descent, such as a soft decision tree.

Three further rules let this work without changing the Runner. First, library versions ship different defaults, just as frameworks do: before scikit-learn 0.22 a random forest defaulted to 10 trees and an RBF SVC to $\gamma = 1/n_{\text{features}}$, and a paper that relied on defaults relied on those. The Coder sets the paper-era values explicitly and discloses each one. Second, the command-line contract every generated script must satisfy is unchanged: flags with no meaning for an estimator, such as a learning rate for a random forest, are accepted as documented no-ops, while the sample caps that make the cheap stages cheap still apply, and loss and epoch fields in the metrics file may be empty. Third, when a claim is an average over repeated runs or cross-validation folds, the full run performs every repetition and reports the mean alongside the per-run values, while the cheap stages run a single repetition.

3.6 Runner changes

The Runner’s interface did not change: it still invokes only the generated `reproduce.sh` and one of its four modes, which is what allowed new task types and model families to be added without modifying its code. Classical estimators needed no Runner change at all. Its image gained pinned `pandas`, `scikit-learn` and `scipy`, installed in the same `pip` command as the pinned `numpy`, so that a transitive upgrade fails the build rather than silently moving a numerical dependency.

4 Prompt Hardening from Real Failures

Running the generalized pipeline on real papers exposed six failures in the prompts of the Coder and of the Runner’s triage step. None was found by inspection; each surfaced in a run. Table 4 lists them with the rule that now addresses each. Every triage

rule was verified by re-running triage on the real log that exposed the problem.

Two of these generalize beyond this project, and are drawn out in Section 7. Rows 1 and 5 share a pattern: in both, a model supplied an identifier from memory that looked exactly as plausible as a correct one. Row 6 is the converse, and the more uncomfortable: the defect was in our own prompt, whose descriptive wording the model implemented literally.

5 Results

5.1 Extraction on the new papers

The Reader ran unchanged on all five papers. Table 5 compares them with the four CIFAR-10 papers it had already processed. Every targeted claim was extracted, as was Tang’s distinction between the L1- and L2-SVM gradients.

Two of the classical-ML papers exposed an assumption built into the Reader. Its claims extractor keeps only a paper’s own results and discards baseline rows, which is the right rule for a paper that proposes a method. Fashion-MNIST proposes a dataset: every row of its benchmark table is a standard classifier. On the first pass the extractor considered 124 candidate rows and kept none, and the architecture extractor found no components. The validator flagged the empty claims against the paper’s table, and the retry kept 26 — the best configuration of each classifier on each dataset. The loop recovered only because the validator noticed: a kind of paper the extractor’s instructions never anticipated produced a well-formed, empty result. The same two papers were, however, the first whose validation converged with no flags remaining. For the SVM guide, the Reader took the rounded 96.9% from the paper’s summary table rather than the appendix’s 96.875%, and did not record the appendix’s accuracies under default parameters as claims.

Three weaknesses surfaced on the neural-network papers, all recorded as open defects. First, validation still does not converge there: both end with five unresolved flags. Second, the same result is extracted two or three times when a paper states it in prose, a table and a figure; Wijaya’s 14 claims describe eight distinct results, at mixed precision (0.911 and 0.91). A Critic comparing against claims will need them deduplicated. Third, and most consequential, the Reader records what a paper leaves unstated only for its *architecture*. Tang’s paper states “stochastic gradient descent with momentum” without a momentum value, and an L2-SVM objective without its penalty constant C . Neither omission was flagged anywhere — and they are exactly the two values that collapsed training (Section 5.3).

Table 4: Prompt failures observed in real runs, in the order they occurred, and the rule that now addresses each. Rows 1–5 arose from the first tabular run; row 6 from its rerun.

	Observed failure	Consequence	Rule now in the prompt
1	The Coder fetched Boston Housing from a remembered CMU StatLib URL, which now returns HTTP 403 to every client	The first execution stage failed in 5 s	Coder: OpenML by identifier first; a direct URL only for data found nowhere else
2	Triage classified the 403 as environment_error	The Orchestrator stopped with zero retries on a failure one regeneration would fix	Triage: a refusal from one source is recoverable; only a container with no network at all is environmental
3	A re-triage suggested switching to “a standard alternative regression dataset”	The fix would change what is being reproduced	Never change what is being reproduced
4	A re-triage suggested load_boston	That loader was removed from scikit-learn in version 1.2	Never suggest an API or source not certainly current; give one fix, not a menu
5	A re-triage named Boston Housing as OpenML data_id=506	Identifier 506 is a different dataset (Boston Housing is 531); training would silently proceed on it	Name no identifier absent from the log; the Coder asserts the dataset’s name and shape in code
6	The Coder’s prompt twice said “OpenML data_id and version”; the script passed both	fetch_openml rejects that combination with a ValueError	Pin by identifier alone, or by name together with a version

Table 5: Reader output across nine papers; the first four rows are CIFAR-10 papers processed before this phase. *Comp.* is architecture components; *own/eq* is equations marked as the paper’s own over equations captured; *Gaps* is **unstated_details** entries; *Flags* is unresolved validation flags after the three-pass cap.

Paper	Claims	Hyp.	Data	Comp.	own/eq	Gaps	Flags
Network In Network	12	17	4	7	1/3	12	4
All Convolutional Net	17	13	4	10	2/6	11	5
Deep Residual Learning	66	28	6	10	2/3	7	8
Wide Residual Networks	61	20	4	11	1/1	11	6
Tang (2013), DLSVM	15	23	3	10	4/11	12	5
Wijaya (2023), housing	14	10	1	8	2/2	8	5
Hsu et al. (2003), SVM guide	3	19	8	2	0/6	5	0
Xiao et al. (2017), Fashion-MNIST	26	14	1	0	0/0	3	0
Frosst and Hinton (2017), soft tree	8	11	3	4	5/5	10	4

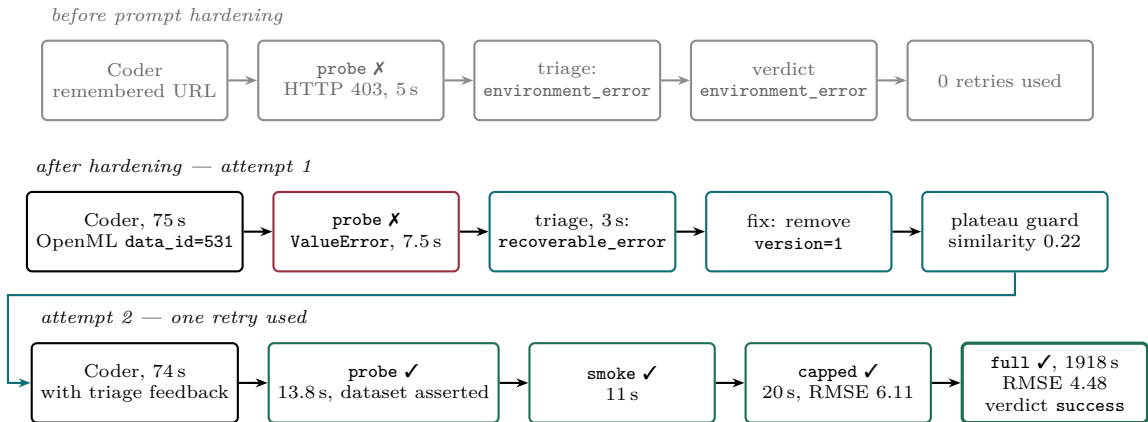


Figure 2: Wijaya’s orchestrated runs. Top: before the prompt changes of Section 4, a dead URL was classified as an environmental fault and the loop stopped without retrying. Middle and bottom: after them, a genuine defect in the regenerated script was triaged correctly, the Coder regenerated with the one-line fix as feedback, and the retry passed every stage. The plateau guard’s 0.22 is far below its 0.98 stopping threshold — and shows how much of the script changed to remove one argument.

5.2 The retry loop repairs real defects

Until this phase, the retry loop had been demonstrated only against a deliberately reintroduced fault: a learning-rate scheduler reading an attribute its constructor never set, which is valid Python and fails only at run time. Wijaya’s two orchestrated runs, shown in Figure 2, supply the first naturally occurring case — and, in the row before it, the failure that motivated the triage rules of Section 4.

Before the prompts were hardened, the first execution stage failed on the dead URL, triage called it environmental, and the loop stopped as designed for an environmental fault — correctly given its input, and wrongly in fact. After hardening, the regenerated script fetched the right dataset but passed both an identifier and a version to `fetch_openml`, which raised. Triage classified the failure as recoverable and proposed removing the `version` argument, which matches the traceback exactly. The Coder regenerated with that feedback, the plateau guard measured a similarity of 0.22, and the second attempt asserted the dataset’s name and shape and passed every stage through `full`. The whole repair cost two generation calls, one triage call, and under three minutes.

The classical-ML and soft-tree runs added two more naturally occurring repairs. The SVM script’s cheap stages trained on the first 256 training rows; `svmguide1`’s training file is sorted by label, and its first 2,000 rows all belong to one class, so scikit-learn refused to fit. Triage classified the failure as recoverable, and the regenerated script sampled the cheap subset at random and passed every stage. The soft-tree script modified a tensor in place inside the autograd graph, a classic PyTorch error that surfaces only in the backward pass; triage identified the in-place operation, and the regenerated script ran. The plateau guard measured similarities of 0.25 and 0.27 for the two regenerations.

5.3 The first full run, and its collapse

Tang’s generated script passed `probe`, `smoke` and `capped` with training error falling from 75% to 31% — evidence, as far as those stages can provide it, that the network learned. Its `full` run was ReproBot’s first at a paper’s own scale. By epoch 20 its training error had risen to 83%, and it then settled near 89.5% (Figure 3). It was noticed by a team member reading the live training log. No stage noticed, and had the run completed, the Runner would have reported *success*: the script exited normally and wrote a well-formed metrics file.

Diagnosis. For a one-vs-rest squared hinge loss, a network that ignores its input can do no better than a constant score per class, $s_k = 2p_k - 1$, where p_k is the frequency of class k . The resulting loss is $\sum_k 4p_k(1 - p_k)$, and the best such network always predicts the most frequent digit. Evaluated at MNIST’s class frequencies, both quantities match

Table 6: The collapsed run against a network that ignores its input. The small excess is the L2 penalty and the input noise.

	Input-ignoring optimum	Observed
Training loss	3.599	~3.607
Training error	88.8%	~89.5%

the observed run (Table 6). The network had died: early oversized updates drove every hidden unit’s pre-activation negative, so every ReLU output zero for every input and no gradient reached the hidden layers; only the output bias still learned, to exactly the input-ignoring optimum. A decaying learning rate cannot recover from this, since a zero gradient stays zero at any step size.

Ablation. The Coder had disclosed four values the paper does not state: momentum 0.9, $C = 1.0$, PyTorch’s default initialization, and unwhitened PCA features. To isolate the cause, we wrote a harness that imports the model, loss and data loading *from the generated script itself*, holds every paper-stated value fixed, and varies one of the four guesses at a time for 30 epochs of the paper’s schedule, measuring the fraction of hidden units that output zero for every one of 2,000 probe images. Table 7 reports the six configurations.

The collapse requires momentum 0.9 and $C = 1.0$ *together*: reducing either alone stabilizes training. Momentum 0.9 amplifies the effective step roughly tenfold, and $C = 1.0$ scales the squared-hinge gradient tenfold relative to $C = 0.1$. Each value is a reasonable default in isolation; the pair is too aggressive for this loss at the paper’s learning rate of 0.1. Whitening the PCA features enlarges the inputs — the mean squared norm rises from 46 to 70 — and collapses fastest, which supports the Coder’s literal, unwhitened reading. Configuration A degraded but did not fully collapse within 30 epochs while the real run did; the two used different random streams for the input noise, so the collapse is seed-dependent, which is precisely what makes it hard to catch with short runs.

We selected $C = 0.1$ (configuration D) as a one-line change to the existing script rather than a regeneration, so that nothing else could change at the same time. It keeps the momentum consistent with the paper’s “with momentum”, produced the healthiest network, and concerns a constant the paper itself treats as tuned elsewhere.

5.4 First fidelity measurements

All five papers completed their check stages, and four completed `full` runs at their papers’ stated settings. Table 8 reports the results against the published claims.

Tang. With $C = 0.1$, training error fell smoothly to 0.062% with no sign of collapse (Figure 3), and

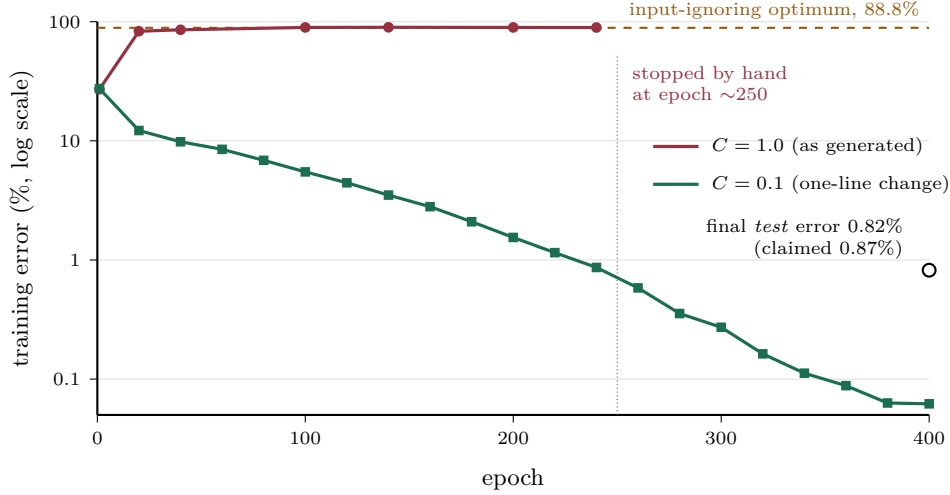


Figure 3: Training error for Tang’s reproduction at the paper’s full setting, from the run logs. As generated ($C = 1.0$, momentum 0.9), the network learned for a few epochs and then collapsed onto the error of a network that ignores its input; the run was stopped by hand. With the single change $C = 0.1$, training error fell smoothly to 0.062%, and the test set, evaluated once after the final epoch, gave 0.82% against the claimed 0.87%. The loss values of the two runs are not comparable, since C scales the loss, so error is plotted.

Table 7: Ablation over the four unstated values, each changed alone from the generated configuration (A). *Dead* is the fraction of second-layer hidden units that output zero for all 2,000 probe images at epoch 30; first-layer units were never dead in any configuration.

Configuration		Test @5	Test @30	Dead @30	Outcome
A	As generated (mom. 0.9, $C = 1.0$)	6.6%	47.1%	0.71	unstable
B	Momentum 0.5	4.0%	2.62%	0.38	stable
C	Momentum 0.0	4.2%	2.66%	0.14	stable
D	$C = 0.1$	4.4%	2.80%	0.04	stable, healthiest
E	PCA whitened	79.8%	90.3%	0.996	total collapse
F	Init. $\mathcal{N}(0, 0.01)$	5.0%	4.2%	0.31	stable, slower

Table 8: ReproBot’s first full-fidelity measurements, compared against the claims by hand. Runtimes are the full stage’s wall clock on the development CPU; some runs shared it.

Paper	Claim	Claimed	Reproduced	Runtime	Reading
Tang (2013)	MNIST test error	0.87%	0.82%	3,223 s	Consistent with the claim, after one unstated value was corrected by hand
Wijaya (2023)	Test RMSE	3.02	4.48	1,918 s	Not reproduced on this split
Wijaya (2023)	Train RMSE	2.69	2.94	—	Close; see text
Hsu et al. (2003)	svmguide1 test accuracy	96.9%	96.63%	73 s	Implementation exact; the grid search chose different C , γ
Xiao et al. (2017)	Random forest test accuracy, mean of 5	0.873	0.8773	200 s	Close; all five runs above the claim
Frosst and Hinton (2017)	MNIST test accuracy	94.45%	—	—	Did not learn; check stages passed regardless

the test set, evaluated once after the final epoch, gave 0.82%. The difference from the claim is five test images out of 10,000. With a single seed we cannot measure run-to-run variance; as a reference scale, the binomial standard error of a 0.85% error rate over 10,000 samples is about 0.09 percentage points. We therefore read the result as consistent with the claim, not as an improvement on it.

Two qualifications matter for how this result may be cited. It is not autonomous: a human chose C . And although the ablation selected on network health — dead units and distance from the input-ignoring loss — it also logged short-horizon test error, so the choice was not made blind to the test set. The accurate statement is that *the generated implementation replicates the paper’s claim once one hyperparameter the paper omits is corrected*.

Wijaya. The reproduced test RMSE of 4.48 is 48% above the claim, while the training RMSE of 2.94 is within 9% of it. The training log qualifies the headline number (Figure 4). The generated script, following the paper’s description, held out 20% of the training rows for monitoring. On those 81 rows, which the model never trained on, the RMSE stayed between 3.2 and 3.5 for most of training and ended at 3.31 — close to the claimed test value. Only the 101-row test set produced 4.48. Two held-out subsets of the same 506-row dataset thus disagree by more than the gap being measured.

The paper states its split sizes but not how rows were assigned, so the script used a seeded random split. On a dataset this small, with a target censored at its maximum value, which rows fall into a 101-row test set plausibly dominates the result. A single seed cannot separate an unfaithful implementation from an unlucky split, and we do not claim either. Two further candidates remain untested: the Coder did not reproduce Keras’s batch-normalization and initialization defaults (Section 7), and a constant learning rate left the final epoch’s training loss noisy.

SVM guide. ReproBot reproduced the paper’s whole procedure: scale each feature to $[-1, 1]$ using the training set’s ranges, evaluate 110 (C, γ) pairs by five-fold cross-validation, retrain on the full training set with the best pair, and test once. It reached 96.625% against the paper’s 96.875%, a difference of 10 of the 4,000 test instances. The cause is identifiable. The search selected $C = \gamma = 8$ at a cross-validation accuracy of 96.99%, where the paper’s search selected $C = \gamma = 2$ at 96.89%. The two candidates are within a tenth of a point of each other, and which one wins depends on how instances are assigned to folds, which the paper leaves to its tool’s random shuffle. To separate the implementation from the search, we evaluated scikit-learn’s **SVC** directly at the paper’s settings, outside the pipeline. It reproduced all three of the paper’s reported test accuracies exactly: 66.925% on unscaled data with default parameters, 96.15% on scaled data with defaults, and 96.875% at $C = \gamma = 2$. Since **SVC** wraps

LIBSVM this is expected, but it establishes that the reproduction is exact in its implementation and differs only in one stochastic step of its protocol.

Fashion-MNIST random forest. The five runs gave between 0.8753 and 0.8792, with a mean of 0.8773 against the claimed 0.873. All five exceed the claim, so the gap of 0.4 points is not run-to-run noise. Tree ensembles are insensitive to monotonic rescaling of pixel values, so the unstated preprocessing is an unlikely cause; the scikit-learn version the paper used, also unstated, is the more plausible one, and we did not isolate it. We read this as a close rather than an exact reproduction.

Soft decision tree. The generated script had to guess a dozen values the paper does not state, and disclosed each. After the retry repaired its crash, its three check stages all passed, yet its accuracy stayed between 5% and 9%, below the 10% of random guessing, and its training loss was negative and identical to four decimal places across every epoch. The cause is instructive. The paper prints its loss as $L(\mathbf{x}) = -\log(\sum_{\ell} P^{\ell}(\mathbf{x}) \sum_k T_k \log Q_k^{\ell})$, but the bracketed quantity is a weighted log-probability and therefore never positive, so its logarithm is undefined; the evident intent is the expected cross-entropy $-\sum_{\ell} P^{\ell}(\mathbf{x}) \sum_k T_k \log Q_k^{\ell}$. The Reader extracted the equation verbatim and marked it as the paper’s own, and the Coder implemented it faithfully, as its rules require — negating the bracket to keep the logarithm defined. The result minimizes the negative log of the cross-entropy, which *maximizes* the cross-entropy: the tree is trained to be wrong. The full run confirmed it, with test accuracy falling to 0.15% and the loss settling at $-\log(20.7)$, where 20.7 is exactly the negative log of the probability floor the script clamps to. Replacing the one line with the expected cross-entropy, outside the pipeline, produced a model that learned immediately: 54% test accuracy after three epochs on 5,000 images. The regeneration also silently changed several guessed values — the learning rate from 0.01 to 0.1, the batch size from 128 to 32 and the penalty strength from 0.1 to 0.01 — none of which the triage feedback had asked for. We report no number for this claim.

5.5 Cost

One Coder call consumed approximately 27,000 input and 9,500 output tokens and took 75 s; one triage call, about 2,400 input and 240 output tokens, in 3 s. Compute again dominated, and in a way worth recording. Both full runs shared the CPU, and each container claimed every core. Wijaya’s 1000 epochs over 324 training rows took 1,906 s — about 1.9 s per epoch of 11 small batches, which is thread contention rather than model cost — and Tang’s run took 54 minutes. The Runner deliberately leaves CPU limits unset, because an arbitrary limit produces failures indistinguishable from crashes; the practical conse-

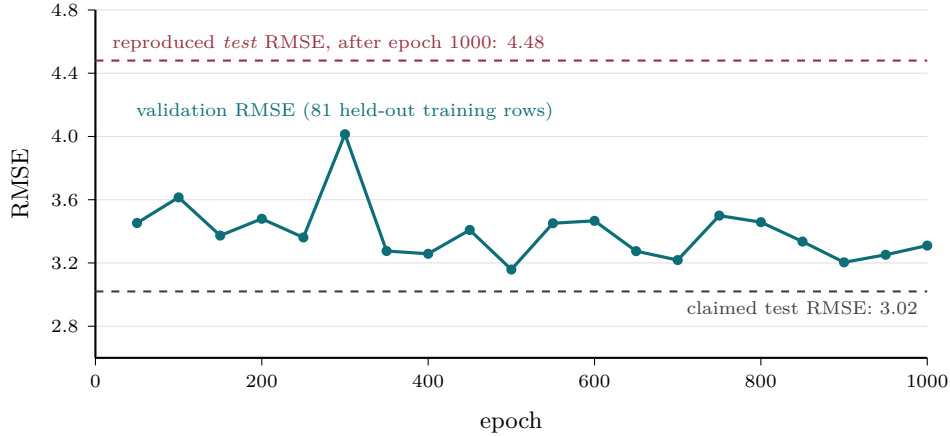


Figure 4: Wijaya’s reproduction at the paper’s full setting, from the run log. The model trained on 324 rows; RMSE on 81 further held-out training rows (every 50 epochs; epoch 1, at 20.7, is off the scale) stayed near the claimed test RMSE throughout, while the 101-row test set, evaluated once at the end, gave 4.48. Two held-out subsets of the same 506-row dataset disagree by more than the gap being measured, which is why a single split cannot adjudicate this reproduction.

quence is that concurrent runs should be avoided.

6 A Front End: The Pipeline Viewer

In parallel with the pipeline work, the team built ReproBot’s first user interface: `viewer/`, a dashboard written in Streamlit, developed on a feature branch. The original proposal planned a Gradio demonstration for the final month; the viewer is an early step towards it, and is already useful as a working tool.

What it shows. The main page lists every paper with a status table — whether OCR, Reader and Coder output exist, and how many validation flags remain. Selecting a paper opens tabs for its method summary, architecture notes, claims, hyperparameters, data pipeline, validation flags, raw extraction JSON, generated training script with its `reproduce.sh`, and the raw OCR Markdown. A reviewer can therefore inspect what the Reader extracted and what the Coder did with it side by side, without opening files.

What it can run. An “Import a paper” section accepts a PDF upload, with an optional page limit so that a new paper can be tested cheaply before committing to the whole document. Per-paper buttons then run Reader extraction or code generation. Two design decisions make this safe. Every action calls the corresponding stage’s own entry point, unmodified, so the viewer cannot drift from the command-line pipeline. And uploaded papers are stored in a separate directory rather than the curated dataset directory, which is maintained by another team member.

Status. The branch was verified with strict type checking, the repository’s pre-commit hooks, and

a headless Streamlit run that rendered every tab against synthetic fixtures. It is not yet merged. Two gaps remain before it is: it does not yet display Runner or Orchestrator output — the results this report is about — and it predates the generalization of the Coder and Runner, so its dependency changes need reconciling with the updated lock file before it can merge. Extending it to show a run’s attempts, triage decisions and metrics, and eventually the Critic’s verdict, is the natural route to the planned demonstration.

7 Engineering Findings

7.1 Where the library is the method, exactness is achievable

The SVM guide shows that some claims can be reproduced exactly, not merely within noise: the same solver on the same data gives the same number to three decimal places. This sharpens what a Critic must do. A claim produced by a deterministic library call admits essentially zero tolerance, and a gap there is diagnostic; a claim produced by stochastic training admits a tolerance set by run-to-run variance; and a claim like the SVM guide’s mixes the two, exact in its solver and stochastic in its model selection. One tolerance for all claims would be wrong for most of them.

7.2 The escalation ladder does not fit models without epochs

For the random forest and the SVM, `smoke` and `capped` produced identical numbers — 0.7891 and 94.92% respectively — because the two stages differ only in the number of epochs, which neither model has. For non-iterative models the fourth stage adds

cost and no information; a ladder over training-set size would fit them better.

7.3 Execution success is not evidence of a result

Static checks cannot establish that generated code runs; the injected scheduler fault above parsed cleanly and failed only in execution. This phase established the next step: that code running to completion cannot establish that it worked. Tang’s collapsed run would have exited normally and been reported as a success. The escalation ladder could not have caught it — `capped` trains for about fifteen steps, and the collapse needed thousands — and `capped`’s stated purpose, checking that training learns, is not enforced by anything: the Runner reads only the exit code.

The soft decision tree repeated the lesson with no scale involved: all three check stages passed with accuracy below chance and a constant, negative loss. The same pattern appeared at smaller scale throughout the phase. The `full` mode of every generated `reproduce.sh` wrote its metrics to a different filename from the one the Runner reads, so both fidelity results above reached the Runner only through its fallback parser; this is now fixed. The OCR, Reader and Coder stages exit with status zero even when every paper fails, which is how a missing image-library dependency went unnoticed on the first OCR attempt of the phase. Every such seam is a place where failure can look like success.

7.4 Framework defaults are part of the method

Two findings point the same way. HuggingFace `Trainer`’s default gradient clipping made a script look stable that, written faithfully, diverges. And a paper written in Keras inherits Keras’s defaults wherever it is silent, so reproducing it faithfully in another framework means reproducing those defaults. The Coder was instructed to do so, and followed the instruction only partly: Wijaya’s script matched Keras’s Adam ϵ but used PyTorch’s batch-normalization momentum and ϵ and PyTorch’s default initialization, and disclosed the choice rather than following the rule. Disclosure makes a deviation visible; it does not make it faithful.

7.5 A remembered identifier is worse than a missing one

A model that lacks an identifier fails visibly. A model that recalls one wrongly — a dead URL, a removed loader, a dataset number one digit away from correct — produces an artifact that looks correct and, for a dataset identifier, trains without complaint on the wrong data. The defence that worked was not asking the model to be more careful, but removing the

opportunity: triage may not name identifiers absent from the log, and the generated script must verify what it fetched before using it.

7.6 Prompt wording is executed literally

The one defect this phase introduced into generated code came from our own prompt, which described OpenML datasets as pinned by “`data_id` and `version`”. The description was meant loosely; the model passed both arguments. Prompts for code generators are read as specifications, and descriptive phrasing in them needs the same care as an API contract.

7.7 Unstated values fail jointly

The Coder’s disclosure discipline worked as designed: momentum 0.9 and $C = 1.0$ were both listed as guesses. What it could not express was that the *combination* was unsafe. Reporting each unstated value in isolation is necessary but not sufficient; the values that matter most may be the ones that interact, and finding that out required an ablation. This argues for recording unstated hyperparameters as carefully as unstated architecture, and for a cheap automated sensitivity probe over them.

7.8 One seed cannot adjudicate a gap

Wijaya’s two held-out subsets disagreed by more than the gap between the reproduction and the claim. A Critic that compares one run against one published number under a fixed tolerance would declare this reproduction a failure — and would declare a lucky split a success. A tolerance must account for the variance of the quantity being compared, which in practice means several seeds per claim.

7.9 Regeneration rewrites, measurably

The retry loop regenerates the whole script on every attempt rather than patching the line at fault. Wijaya’s repair quantifies the cost: triage’s fix was the removal of one keyword argument, and the regenerated script shared only 22% of its lines with its predecessor. The repair succeeded, but almost everything else in the script also changed, unreviewed. The plateau guard’s threshold of 0.98 now has four observations — 0.41 on the injected scheduler fault and 0.22, 0.25 and 0.27 on real ones — and none came near it. The soft tree shows the cost directly: its repair changed three guessed hyperparameters that nobody had asked to change.

8 Development Methodology

ReproBot is built by a human team working with an orchestrating AI coding session, which delegates

Table 9: Status against the project’s original four-month plan. ✓ done; ∼ partial or in a different form; ✗ not started.

Phase	Planned	Status
Month 1	PDF ingestion; Reader schema; shared-memory object	✓ Complete; five fields
Month 2	Coder and Docker sandbox for a pilot set; Orchestrator skeleton with a single hard-coded retry	✓ Exceeded: a bounded retry loop with triage-driven feedback
Month 3	Critic verdict logic	✗ Not started; first real inputs now exist
	Real retry loop with targeted feedback	✓ Repaired three real defects
	Report Generator	✗ Not started
	Single-shot vs. iterative ablation	∼ Three observed cases: each fails single-shot and runs after one retry
Month 4	Expansion towards 20 papers	∼ Seven papers across five model families
	Demonstration interface	∼ Streamlit viewer on a feature branch

scoped tasks to specialized subagents — research agents that survey precedent, coding agents that implement and verify a change, review agents that critique an artifact before a costly action, and validation agents that audit output read-only — and logs every delegation with its brief and outcome. Most of this phase’s work was investigative rather than constructive and was carried out directly by the orchestrating session. One scoped implementation — the generalization of the Coder and Runner — was delegated to a coding subagent.

That delegation produced the third instance, across the project, of a subagent’s report being wrong in a way only independent verification caught. The agent reported its *Network In Network* regression test as a clean pass. It had passed: the script executed and the metrics file was well formed. The loss was approximately 947. The discrepancy was found on review, and led directly to the hidden-default finding above. The practice of treating an agent’s report as a claim to check against the diff and the output remains the most useful single habit we have adopted.

The two most consequential decisions of the phase were human: the direction change to support multiple task types, and the choice of C for Tang’s rerun. The collapse itself was spotted by a person reading a log. We state this plainly because an automated replication system is exactly the kind of project in which human intervention can be quietly absorbed into a result.

9 Progress Against the Timeline

The project was planned over four months: the Reader in the first; the Coder, sandbox and a skeleton Orchestrator in the second; the Critic, a real retry loop and the Report Generator in the third; and evaluation, ablation and a demonstration in the fourth. Table 9 compares the work to date with that

plan. The retry loop arrived ahead of plan and in stronger form than planned. The Critic and the Report Generator, both Month 3 deliverables, have not started, and the phase’s effort went instead to generalization and to obtaining the first measurable results — a trade we consider correct, since a Critic without measurable results would have had nothing to judge.

10 Limitations

The fidelity results are compared by hand, mostly from single runs. Only the random forest reports a spread over runs, and no stage of the system performed any comparison.

A misprinted equation defeats faithful implementation. No stage can currently tell an equation the paper printed wrongly from one it printed correctly; the soft-tree failure was diagnosed by hand.

Model-family coverage is still narrow. No gradient-boosting library (XGBoost, LightGBM, CatBoost), sequence model, graph network or text model has yet been run, and one model family, the soft decision tree, produced no result at all.

Tang’s positive result required a human. Tang’s reproduction matches its claim only after a person chose one hyperparameter, informed by an ablation that had logged test error.

The negative result is not attributed. Wijaya’s gap may stem from the unstated split, from framework defaults the Coder did not reproduce, or from the implementation, and one seed cannot distinguish them.

The CIFAR-10 benchmark remains unmeasured. GPU compute is still required for it, and the two CPU-sized papers are not substitutes for the planned evaluation, only a way to test the pipeline’s claim end to end.

The Reader does not record missing hyperparameters, and claims are duplicated across a paper’s prose, tables and figures. Validation still does

not converge, and two of the eight CIFAR-10 papers still cannot be extracted.

A note on the dataset. Boston Housing includes a feature derived from the proportion of Black residents in each town [3], which is among the reasons scikit-learn deprecated and then removed its loader. We used it only because the paper under replication did, and do not recommend it for any other purpose.

11 Next Steps

Build the Critic, with variance in mind. Its verdict should remain explicit arithmetic against a numeric tolerance using the claim’s direction, as planned; Wijaya shows that the tolerance must be informed by several seeds per claim rather than one run. Tang’s collapsed run is its first test case: a verdict of *fail* at 89% error against 0.87% is the minimum it must deliver.

Make the Runner check that training learns. Reading `train_metric` at `capped`, and comparing a run’s loss against the trivial input-ignoring floor, would have flagged Tang’s collapse within twenty epochs. This is smaller than the Critic and catches a different failure.

Record and probe unstated hyperparameters. Extend the Reader’s gap-recording to hyperparameters, and automate what the Tang ablation did by hand: short runs over candidate values for the unstated ones, selected on training health and a validation split — never the test set.

Patch rather than regenerate. A line-range edit, as in AutoReproduce [9], would keep a one-argument fix to one argument.

Fit the escalation ladder to the model. Scale training-set size rather than epochs for non-iterative models, and reject a check stage whose loss does not move.

Widen coverage. Run the diversity targets already selected — gradient-boosting libraries, a graph convolutional network, a text CNN and a temporal convolutional network — which requires adding XGBoost, LightGBM and CatBoost to the sandbox image.

Merge and extend the viewer to show Runner, Orchestrator and Critic output, as the basis of the final demonstration.

Secure GPU compute for the CIFAR-10 set, and build the Report Generator and a single entry point from PDF to report, after fixing the stages that exit successfully on failure.

12 Conclusion

This phase produced ReproBot’s first replication measurements, across five papers and five model families. Three reproductions land close to their claims, one of them exact in its implementation; one

misses its claim by a margin a single split cannot explain; and one never learned. Getting there required generalizing the pipeline beyond image classification and beyond neural networks, turning six real prompt failures into explicit rules, and diagnosing a training collapse that every stage of the pipeline reported as a success.

That last point is the phase’s main result, and it happened twice. ReproBot can now read a paper, write an implementation in the right library, execute it at the paper’s own scale, and repair its own execution failures. What it cannot yet do is tell whether the number it produced is right. Every measurement in this report was judged by people, and one of them needed a person to fix it first. Building the component that makes those judgements automatically — with a tolerance that respects whether a claim is deterministic or stochastic — is the work of the next phase.

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